



EIA CASE STUDY ON:

SABARIGIRI HYDROELECTRIC POWER PROJECT

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INTRODUCTION

- Sabarigiri Hydroelectric Power Project is the second-largest hydroelectric project in Kerala.
- Established in 1966, plays a crucial role in the state's power generation.
- Key contributor to renewable energy and sustainable power.
- Significantly reduces reliance on non-renewable energy sources.
- Located in the Western Ghats, utilizing the region's water resources efficiently.

PROJECT OVERVIEW

- Location: Pathanamthitta District, Kerala
- Commissioned: 1966
- Installed Capacity: 340 MW

Key Features:

- Kakki and Anathode Dams: Collects and regulates the water needed for power generation.
- Powerhouse at Moozhiyar: Electricity generation takes place here, using cutting-edge hydropower technology.
- 6 Francis Turbines: Equipped with six high-efficiency Francis turbines, with the ability to handle large water flows, enabling continuous electricity production.

PROJECT BACKGROUND

- **Location:** Situated in the Pathanamthitta district of Kerala, India, near the Pamba River.
- **River Basin:** Utilizes the water from the Pamba River, primarily from its tributaries.
- **Capacity:** Initially commissioned with an installed capacity of 300 MW, later upgraded to 340 MW.
- **Inception:** The project was commissioned in 1966 as part of Kerala's effort to boost its power generation capacity.
- **Type of Plant:** It is a conventional hydroelectric power station, where electricity is generated using the flow of water through turbines.

PROJECT BACKGROUND

- **Dams:** The Sabarigiri project includes multiple dams such as the Kakki and Anathode dams, which create a reservoir for storing water.
- **Water Storage:** The plant relies on water storage in Kakki reservoir, from where water is channeled to generate electricity.
- **Operational Agency:** Managed and operated by the Kerala State Electricity Board (KSEB).
- **Transmission:** Electricity generated from the plant is transmitted to various parts of Kerala via a network of transmission lines.
- **Importance:** It is the second-largest hydroelectric project in Kerala and contributes significantly to the state's electricity supply.

ENVIRONMENTAL IMPACTS

- **Deforestation:** Large areas of forest cleared for dam construction and reservoir formation, leading to loss of biodiversity.
- **Wildlife Displacement:** Animal habitats affected, causing displacement of species and disruption of ecosystems.
- **Water Flow:** Alteration of natural river flow modified, impacting aquatic life and downstream water availability.

ENVIRONMENTAL IMPACTS

- **Soil Erosion:** Changes in land use and deforestation lead to increased soil erosion in surrounding areas.
- **Sedimentation:** Reservoirs cause sediment accumulation, affecting water quality and reducing storage capacity.
- **Climate Impact:** Waterlogged vegetation in reservoirs can release methane, a potent greenhouse gas.

SOCIAL IMPACTS

- **Displacement of Communities:** The construction of the project led to the displacement of local populations, affecting their homes and livelihoods.
- **Improved Infrastructure:** The project contributed to enhanced local infrastructure, including roads, schools, and healthcare facilities, benefiting the surrounding communities.
- **Economic Opportunities:** It created job opportunities during construction and operation, contributing to the economic development of the region.

SOCIAL IMPACTS

- **Changes in Land Use:** The project altered local land use patterns, affecting agriculture and access to natural resources for nearby communities.
- **Health Impacts:** Increased construction activities and environmental changes led to concerns about air quality and public health in the surrounding areas.
- **Access to Electricity:** The project improved access to electricity in the region, contributing to better living standards and industrial growth.

OPERATIONAL CHALLENGES

- **Aging Infrastructure:** The project was commissioned in 1966, and the aging infrastructure reduces its operational efficiency. It requires constant maintenance and upgrades to keep the power generation steady.
- **Sedimentation in Reservoirs:** Over the years, sediment accumulation in the Kakki and Anathode reservoirs has decreased their storage capacity, affecting both water availability and the efficiency of power generation.
- **Environmental Degradation:** The project has led to the submergence of large areas of forest land, resulting in biodiversity loss, which further complicates the plant's operations by invoking environmental scrutiny and the need for mitigation measures.

OPERATIONAL CHALLENGES

- **Climate Change and Water Availability:** The changing climate is leading to unpredictable water flow, impacting the plant's ability to generate power consistently throughout the year.
- **Erosion and Landslides:** Erosion around the reservoir areas and occasional landslides pose a significant threat to the dam's structure and the overall safety of the project.
- **Community Displacement and Social Challenges:** The creation of reservoirs displaced local communities, and ongoing social issues remain a challenge as the project seeks to balance operations with community concerns and regulatory obligations.

SWOT ANALYSIS

S - Strength

1. Renewable Energy Generation

- Sabarigiri makes a substantial contribution to Kerala's renewable energy supply by using the Pamba River's flow to create electricity without releasing greenhouse emissions.
- It provides a sustainable, long-term energy source that reduces dependence on fossil fuels and mitigates pollution.

2. Flood Control

- helps regulate the flow of water, preventing the risks of seasonal flooding in downstream areas, which can safeguard agricultural lands and local communities.
- The reservoirs created by the dams act as a buffer during heavy monsoons, reducing the threat of destructive floods.

SWOT ANALYSIS

W - Weakness

3. Environmental Challenges:

- The creation of reservoirs has led to the submergence of large forested areas, resulting in deforestation and loss of biodiversity.
- Disruption of the natural flow of the Pamba River has affected aquatic ecosystems and reduced the river's natural health.

4. Ageing Infrastructure:

- The Sabarigiri project was commissioned in 1966, making its infrastructure over 50 years old.
- Ageing equipment requires significant maintenance and modernization to maintain efficiency and safety.
- Reduced operational efficiency due to wear and tear, leading to lower electricity generation compared to its peak performance.

SWOT ANALYSIS

O- Opportunities

5. Modernization and Technological Upgrades:

- There is an opportunity to upgrade the ageing turbines, generators, and other equipment, which would improve the plant's efficiency and extend its operational life.
- Adopting new technologies, such as automated systems, can also enhance operational efficiency and reduce maintenance costs.

6. Community Engagement and Development:

- Increased involvement in local communities can result in projects that create jobs, construct infrastructure, and implement social welfare programs, all of which can enhance people's quality of life.
- Communities can be trained and made aware of environmental conservation and sustainable resource usage through initiatives that guarantee long-term cooperation in reducing the ecological influence of the project.

SWOT ANALYSIS

T - Threats

7. Climate change

- Changing weather patterns, such as reduced rainfall or altered monsoon cycles, can affect the water availability in the Pamba River, reducing the project's power generation capacity.
- Increased droughts and unpredictable river flows pose long-term threats to the project's operational viability.

8. Natural Disasters

- The project is located in a region prone to landslides, earthquakes, and heavy rainfall, which can pose serious risks to the integrity of the dams and infrastructure.
- Reservoir-induced seismicity and landslide risks could lead to catastrophic failures if not addressed adequately through regular monitoring and safety measures.

KEY INSIGHTS

- **Environmental Management:** Future projects should prioritize minimizing ecological impacts, such as forest loss and biodiversity disruption, by adopting sustainable practices and continuous environmental monitoring. Prioritize minimizing ecological damage, like forest loss and biodiversity impact, through sustainable practices.
- **Community Engagement:** Early, transparent, and ongoing dialogue with local communities is essential to mitigate the social impact, particularly displacement, and to secure long-term support for the project.
- **Infrastructure Maintenance:** Regular maintenance and timely modernization of aging infrastructure are critical to sustaining operational efficiency and ensuring the long-term viability of such projects.
- **Climate Adaptation:** Hydroelectric projects need to be flexible and adaptable to the effects of climate change, especially changes in water availability, to maintain reliable power generation.

CONCLUSION

The Sabarigiri Hydroelectric Power Project has been a key contributor to Kerala's energy needs since 1966, generating renewable energy while supporting local infrastructure and flood control. However, it has also caused significant environmental challenges, including forest submergence, biodiversity loss, and social displacement. Despite these trade-offs, the project highlights the need for better management practices, modernization, and community engagement to ensure sustainable development. Lessons learned from Sabarigiri, especially regarding environmental protection and infrastructure maintenance, will be crucial for the future of hydroelectric projects.



THANK YOU